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Uncovering Physics-Based Water Production Patterns Through Well and Reservoir Management Process Modeling

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Abstract

Reservoir management focuses on understanding flow processes within the drainage volume of wells, where high velocities and unfavorable mobility ratios often challenge conventional flow assumptions. Even subtle geological or structural features can significantly affect fluid flow, yet these influences are difficult to represent accurately using traditional numerical models. This paper demonstrates that Well and Reservoir Management (WRM) Process Modeling offers a practical and effective alternative for simulating such complex environments. WRM Process Modeling introduces a streamlined, physics-driven approach to flow analysis, prioritizing critical geo-parameters and augmenting field data with physically consistent synthetic data to enable the use of machine learning (ML) algorithms to identify and rank the key parameters influencing well behavior.

A synthetic case study is presented to illustrate the method. The process begins with expert input from WRM practitioners to identify all parameters potentially influencing a well's WOR signature, along with their plausible ranges. Monte Carlo sampling is then applied to these parameters to generate multiple realizations of conceptual reservoir models. These realizations form a catalog of geological scenarios that associate specific combinations of parameters with corresponding WOR and bottomhole pressure signatures. Comparing actual well data with this catalog enables the identification of the most likely flow setup through pattern recognition powered by ML. Once the well setup is established, targeted and optimized remedial actions can be designed to mitigate excessive water production.

WRM Process Modeling facilitates rapid and robust decision-making based on a physics-consistent understanding of well behavior, avoiding unnecessary complexity and reducing human bias. Uncertainty is addressed through ensemble modeling of conceptual reservoir scenarios, eliminating the need for full data assimilation or history matching, and thereby accelerating the evaluation cycle.

With increasing environmental scrutiny on produced water volumes, reservoir management has a growing role in ensuring sustainable oil and gas operations. This methodology supports that goal by strengthening the technical foundation for well intervention decisions aimed at minimizing water production while maintaining operational performance.

Introduction

Effective reservoir management depends on a solid understanding of flow dynamics in the near-wellbore region. However, conventional full field or regional simulation models often struggle to accurately characterize this zone due to steep velocity gradients, unfavorable mobility ratios, and the influence of subtle geological or flow features not captured due to coarse resolution. The near-wellbore area frequently exhibits complex, nonlinear behavior that is difficult to quantify, and it is poorly represented in typically used simulation grids. As a result, more suitable alternatives include data-driven methods or hybrid WRM process modeling approaches that strike a balance between empirical analysis and full-physics representation.

Produced water management is a critical component of effective reservoir management. Trends in water-oil ratio (WOR) often spark discussions around reservoir heterogeneity, fluid mobility, and displacement geometry, factors that significantly influence water management strategies and operational decisions yet are frequently subject to conflicting interpretations. This paper develops a Water and Reservoir Management (WRM) process modeling approach that integrates data-driven insights with physics-based methods to infer the reservoir and well configurations responsible for observed WOR and flowing bottomhole pressure (FBHP) behavior¹. The objective is to equip WRM teams with a decision-making tool that reduces reliance on subjective interpretation and promotes more consistent, evidence-based strategies.

This paper follows the philosophy of *designing fit for purpose models*², which argue that it is best to build faster models tailored to answer business specific questions over building full field detailed models that intend to know every aspect of the reservoir and give answers to every management question. In this sense, the models proposed in this paper use data commonly available in production wells and try to estimate fundamental reservoir properties like porosity and permeability in well nearby volume.

The methodology enables the identification of the fundamental reservoir characteristics within the well drainage area and foresees the expected water-cut evolution and ultimate recovery for the well, without the need for full-field simulation or complex data assimilation workflows. By combining physically consistent synthetic data, advanced simulation techniques, and ML algorithms, the approach provides a technically robust, time-efficient, and operationally practical framework for water control and reservoir surveillance.

Finally, to streamline the application of the proposed approach and facilitate its usability in operational settings, the entire workflow – from preprocessing to forecasting – can be implemented into a scikit-learn pipeline. The raw data preprocessing and features extraction procedures can be modularized as compatible scikit-learn transformers ready to be implemented in the machine learning pipeline, allowing the final user to interact with a fully automated pipeline where raw production data can be directly input and the forecasting results are obtained without manual preprocessing or feature engineering.

By encapsulating the whole methodology within a standardized pipeline, the solution becomes not only reproducible and robust but also readily deployable for field scale applications.

Methodology

WRM Process Modeling combines expert-driven sensitivity analysis, synthetic simulation catalogs, and ML-assisted pattern recognition to characterize the flow in the near-wellbore volume. The steps include (Figure 1):

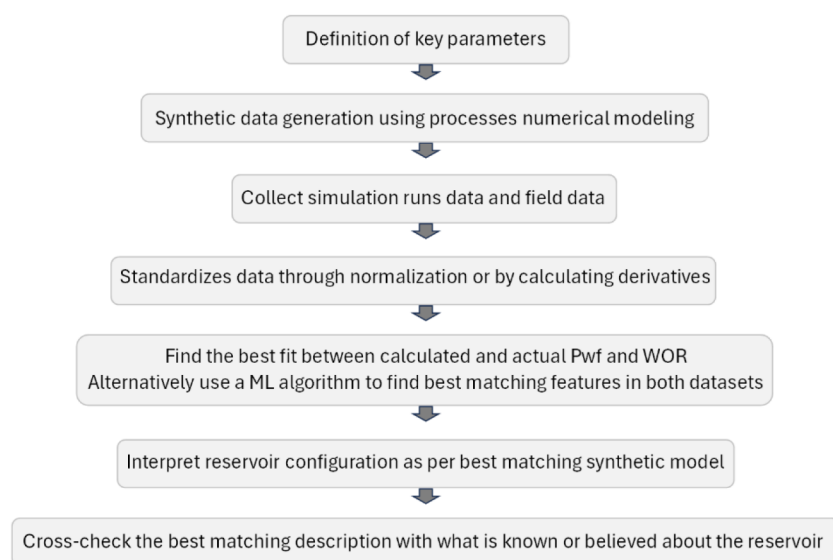


Figure 1—Workflow followed to apply WRM Process Modeling methodology

Definition of key Parameters

Subject matter expertise is used to identify a set of geo-parameters potentially influencing WOR trends. These may include aquifer size and connectivity, rock properties in the well surroundings: porosity, permeability, vertical/horizontal anisotropy, relative permeability curves, and well completion details.

Synthetic Data Generation

Within the WRM Process Modeling framework, thousands of conceptual numerical simulations are generated through Monte Carlo sampling across defined ranges of reservoir and completion parameters. Each simulation produces variable signatures, for this study WOR and FBHP, which are compiled into a catalog of conceptual scenarios. Each scenario represents a physically plausible flow behavior pattern, providing a structured basis for interpreting well performance.

Reservoir heterogeneity can be incorporated through spatial distributions of distinct rock types, while homogeneous cases can be simulated and used as baseline references for comparison.

Collect simulation runs data and field data

Multiple conceptual simulation runs are to be carried out varying the parameters defined previously within acceptable ranges. The input data of those runs as well as the relevant field data profiles (WOR and FBHP) are compiled.

Standardizes data through normalization or by calculating derivatives

Prior to applying ML, it is essential to standardize the data, either through normalization or by calculating derivatives, to ensure that all input variables contribute proportionately to the model training process. Normalization prevents features with larger numerical ranges from dominating those with smaller scales, while derivatives help highlight dynamic trends and reduce the influence of noise or baseline shifts. These preprocessing steps enhance model stability, improve convergence, and increase the reliability of pattern recognition in complex reservoir systems.

Use a machine learning algorithm to find the best matching features

Feature selection and supervised ML (e.g., random forests, gradient boosting) can be applied to the simulation catalog to rank parameters according to their influence on desired outcomes, for instance WOR and FBHP. This reduces dimensionality and focuses the analysis on key drivers. The ML models are tuned using this synthetic set of data.

Pattern Matching and Model Selection

Actual well data are processed by the ML models, enabling efficient identification of the most probable well completion and reservoir (WC&R) configurations. This step replaces subjective interpretation with data-driven diagnostics grounded in physics.

Once the WC&R, well completion and reservoir configuration are identified, appropriate mitigation strategies, such as zonal isolation, conformance control, or shutoff treatments, can be selected and evaluated on a more objective technical basis.

Case example

The methodology is demonstrated using a vertical well completed in a homogeneous reservoir subjected to bottom water drive. The simulations were conducted using ERT³ and a public domain simulator, OPM Flow⁴. A total of 9,000 runs were performed – 7,000 used for training and 2,000 kept for validation – varying the well completion setup, reservoir properties (porosity and permeability, K_v/K_h ratio), water-oil mobility ratio, and the aquifer-to-STOOIP ratio. The grid used was radial with dimensions $40 \times 6 \times 150$ in the r , θ , and z directions, resulting in a total of 36,000 cells. The top 100 layers represent the oil-bearing zone, while the bottom 50 layers represent the underlying aquifer. Input parameters were independently sampled from either uniform or normal distributions, chosen based on the most suitable statistical representation for each variable.

Input Data

The input data used in the numerical simulations is summarized in the Table below (Table 1):

Table 1—Input data used in the numerical simulations distinguishing parameters that were randomly sampled and those that remained constant across all runs.

Parameter	Description	Constant	Random
1 Relative permeability k_{rw} end point	Calculated using the L.E.T. correlation. Parameters of L.E.T. correlation randomly sampled		
2 Porosity	Porosity randomly selected from pdf provided		
3 Permeability	Permeability calculated from porosity using poroperm equation		
4 Relative permeability k_{ro} end point	Kept constant for all scenarios		
5 Top perforation (well completion)	It is always selected as the first (shallowest) cell within the oil-bearing interval of the simulation grid		
6 Bottom perforation (well completion)	Randomly sampled for each simulation run		
7 STOOIP	Modified by changing the count of active radial cells in the grid		
8 Capillary pressure	Kept constant for all scenarios		
9 Drawdown (Reservoir pressure - flowing bottom hole pressure)	Randomly sampled for each simulation and used to compute the liquid production rate		
10 Critical water saturation	Randomly sampled		
11 K_v/K_h	Randomly sampled from pdf supplied		
12 Aquifer size	Modified by randomly selecting multipliers applied to the porous volume of the aquifer's bottom layer		

To ensure the stability of ML algorithms and the relevance of the predictors, it must be verified that the input features are not linear combinations of one another. Redundant information can lead to poorly trained models, complicate interpretation, increase the risk of overfitting, and obscure the physical significance of individual predictors, an especially critical concern in reservoir engineering workflows. To address this, the interdependence among features was evaluated by analyzing pairwise correlations to identify strongly correlated variables.

Figure 2 presents the results of this analysis, showing a collection of scatter plots comparing different pairs of variables. Variables that form a diffuse cloud of points with no discernible trend can be considered independent, while those that display a well-defined curve or linear trend are likely dependent. For instance, the first column at the left-hand side contains several variables plotted against porosity. Most of these

variables appear to be uncorrelated with porosity, indicating independence. However, the eighth plot from the bottom of the column highlights a clear relationship between porosity and STOOIP (stock tank oil originally in place), as expected, since STOOIP is directly influenced by porosity. This confirms the ability of visualization to highlight dependent relationships.

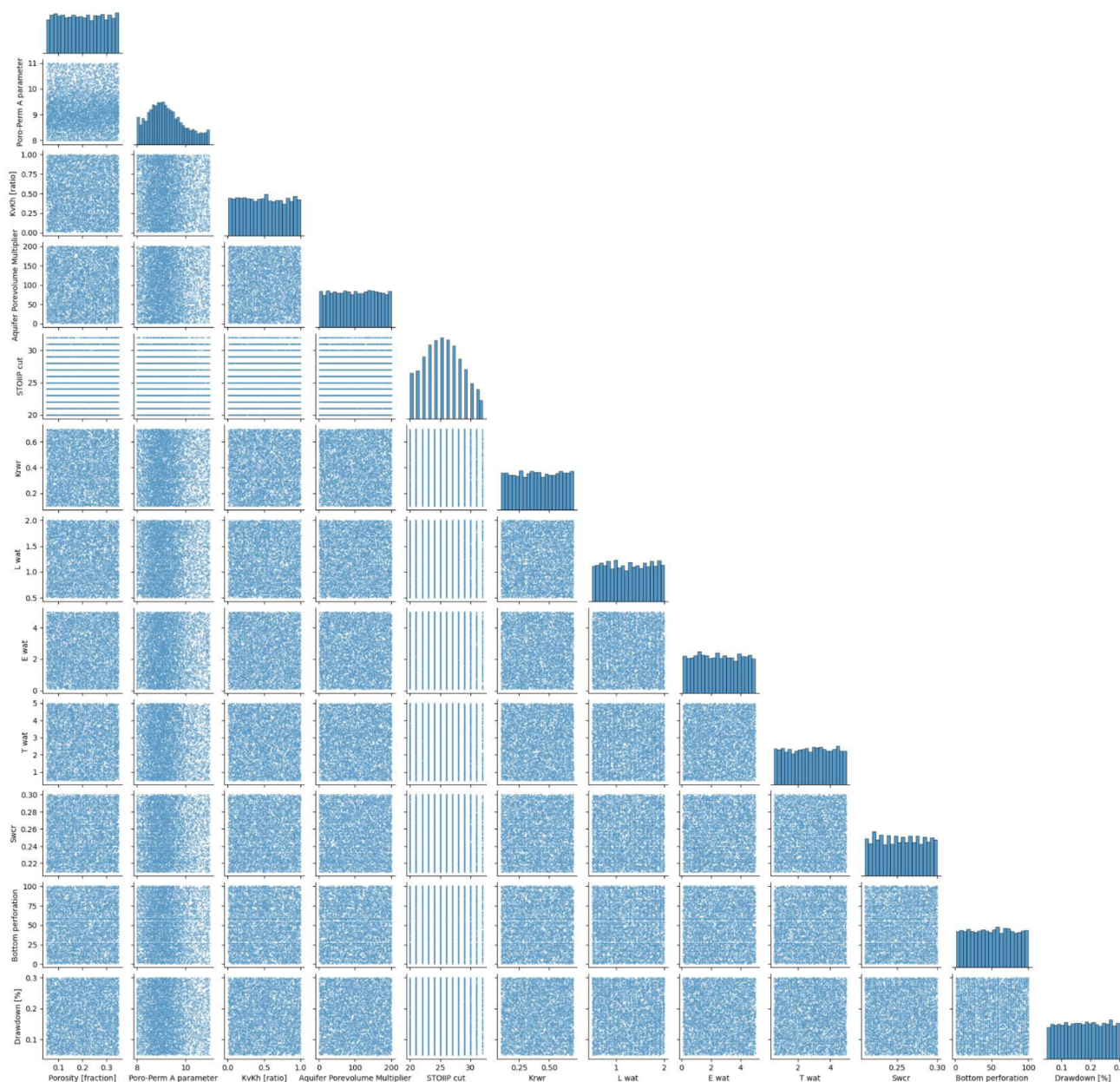


Figure 2—Pairplot of input parameters illustrating their degree of independence. Independent parameter pairs appear as scattered clouds of points, whereas dependent pairs exhibit structured patterns, such as curves or clustered trends.

Some examples of the data input are shown in the Figures below. For example, the porosity-permeability relationship follows an equation of the type of $PERM = 10^{(A * PORO + B)}$ and the outcome is displayed in Figure 3. Note that the spread of permeability increases as porosity increases, indicating greater pore connectivity at larger porosities. The resulting distributions of porosity and permeability are displayed in the histograms in Figure 4.

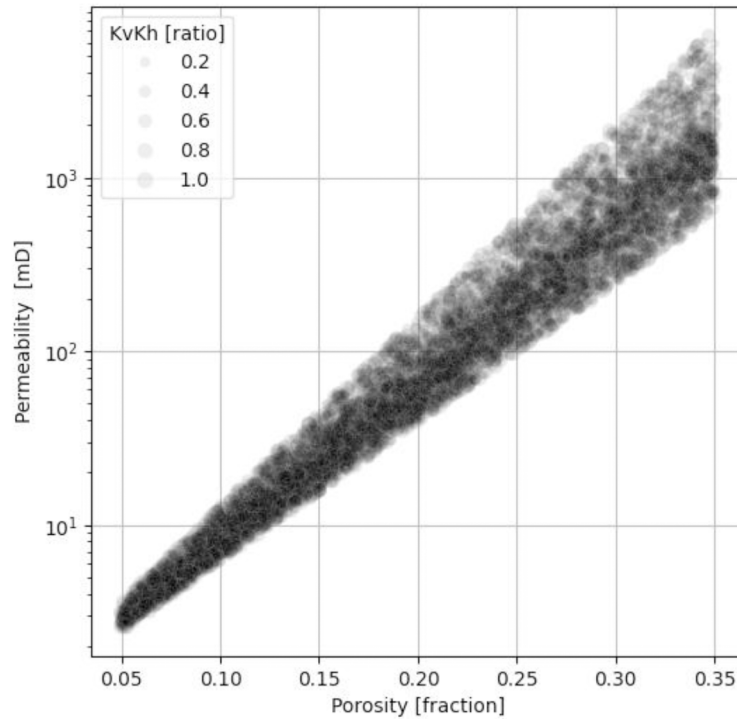


Figure 3—Porosity-Permeability relationship representing the KvKh ratio as bubble size

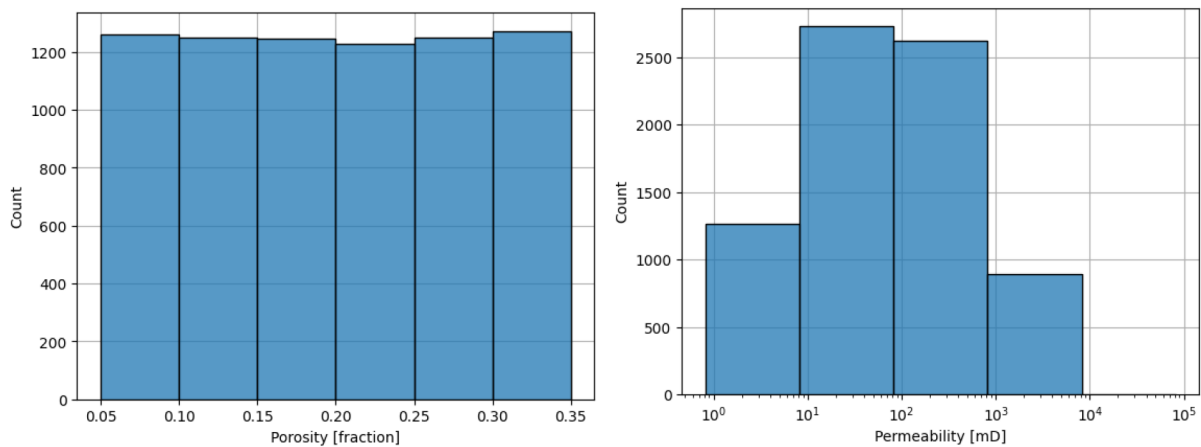


Figure 4—Porosity and permeability distributions used in the numerical simulation

The relative permeability curves were generated using the equation proposed by Lomeland et al.⁵, which consistently produces curves representative of intermediate wettability conditions. In this study, the oil relative permeability curve is held constant across all realizations, while the water relative permeability curve is varied by randomly sampling the parameters of the Lomeland equation. This approach aims to simplify calculations, minimize noise in the input data, and improve the interpretability of synthetic simulation results. Additionally, the saturation endpoints, namely the irreducible water saturation (SWL) and the connate water saturation (SOWCR), are also kept constant throughout the simulations. The resulting relative permeability curves are displayed in Figure 5.

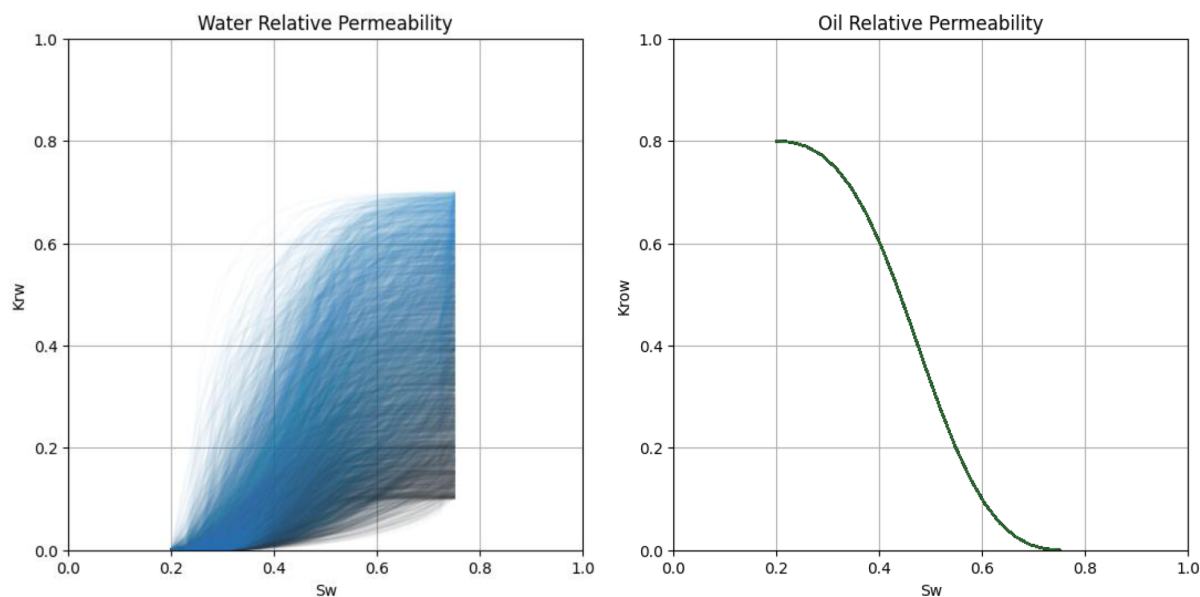


Figure 5—Water and oil relative permeabilities used in the ensemble. Oil relative permeability is kept constant, while water relative permeability is randomly selected

In this study, the mobility ratio is primarily influenced by the water relative permeability, as all other parameters, namely, oil and water viscosities, as well as oil relative permeability end points, are held constant. The oil is assumed to be undersaturated and remains in the undersaturated state throughout the simulation period. PVT properties were derived from the *SPE3CASE1*⁶ dataset, using a solution gas-oil ratio (R_s) of 0.479365 Mscf/stb.

The vertical well completion was varied by specifying the bottommost cell open to flow. The top of the perforated interval was fixed at cell number 1, while the bottom depth was randomly sampled, resulting in the distribution displayed in Figure 6.

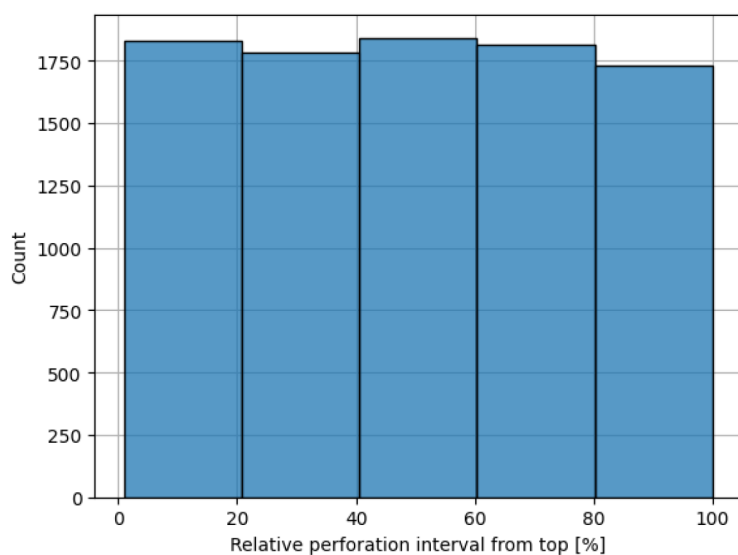


Figure 6—Histogram of the perforated length samples. The top of the perforated interval is always the first (shallowest) cell in the oil-bearing interval, the bottom of the perforated interval is a cell that is randomly selected.

A sample of the output of the numerical simulator is included in Figure 7.

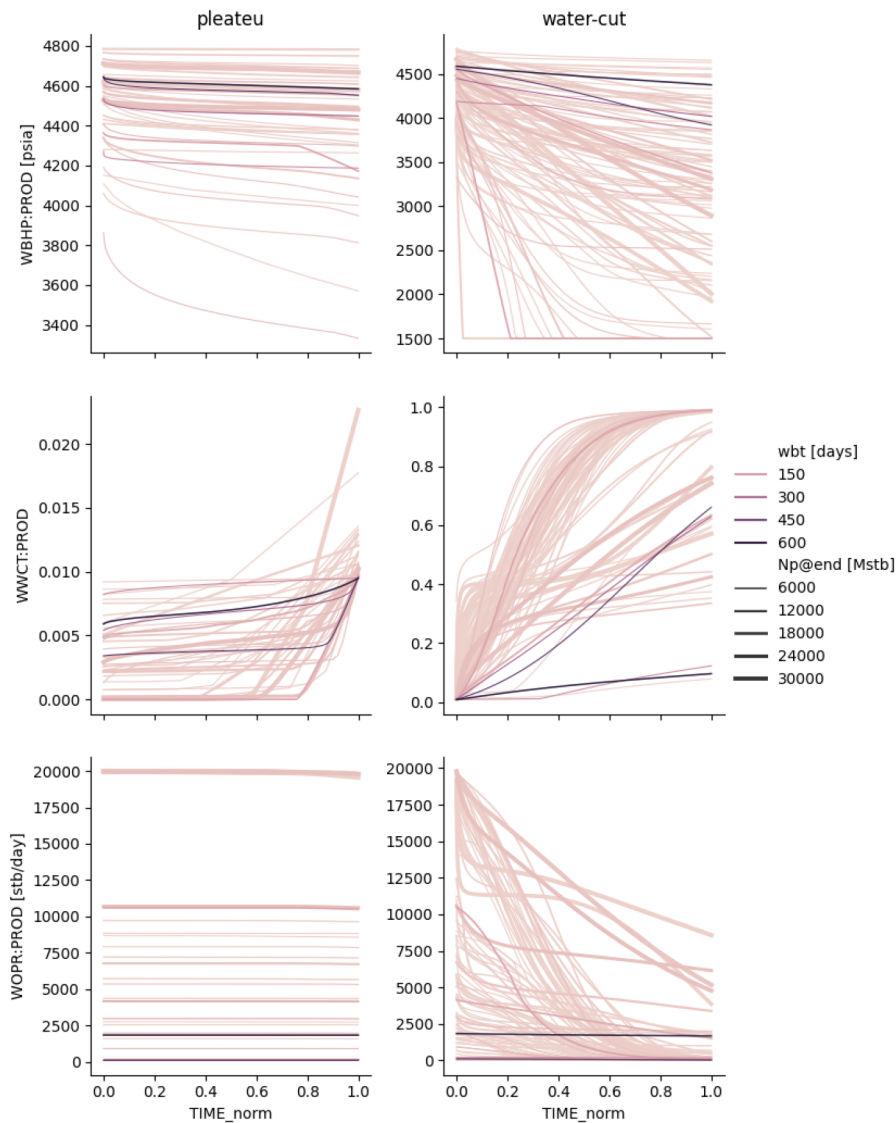


Figure 7—Sample of the output of multiple numerical simulations

The simulations are configured to operate the well at a constant liquid production rate until the water cut exceeds 99%, or until a maximum production period of eight years is reached, whichever occurs first. This production time limit is imposed to avoid indefinitely long simulation runs. Depending on the specific well configuration and reservoir parameters, the well may or may not exhibit an initial period of dry oil production prior to water breakthrough. The production schedule is shown in [Figure 8](#).

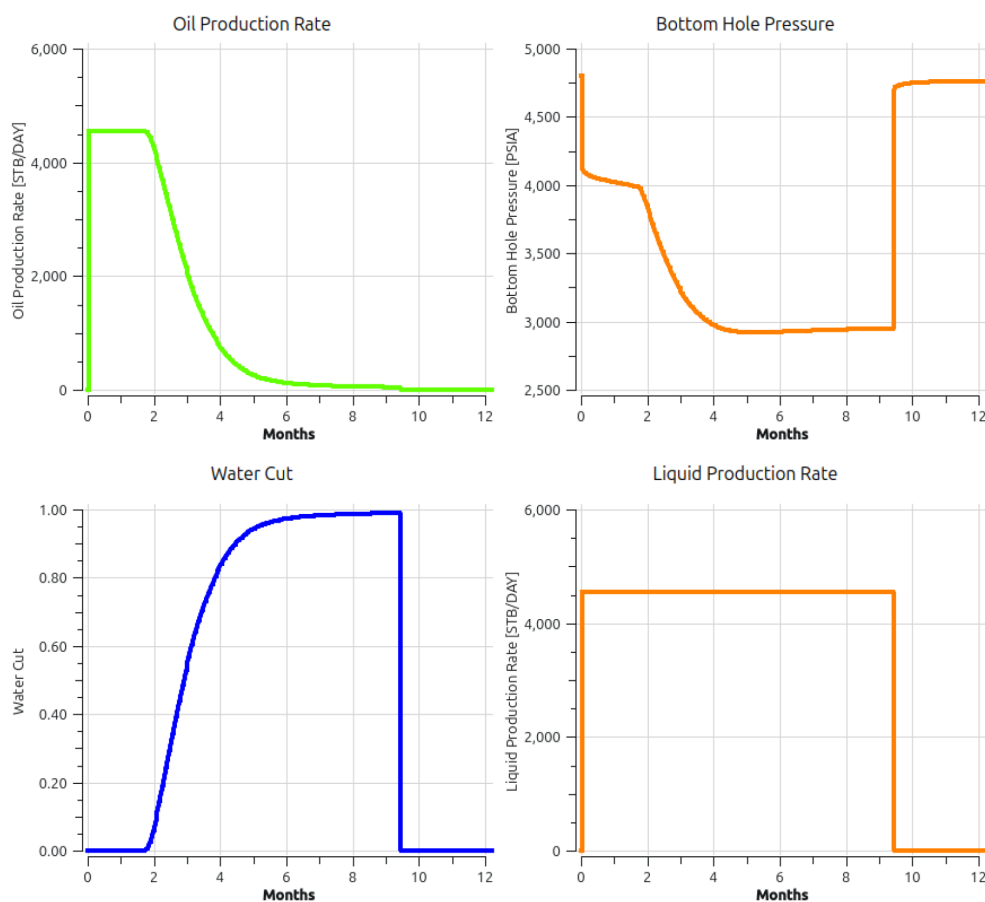


Figure 8—Simulation production schedule: each run terminates when the water cut reaches 99% or after 8 years of production, whichever occurs first.

Reading simulation results

A total of 9,000 runs were carried out with the numerical simulator, each individual run with specific values of porosity, poro-perm coefficients A and B, K_v/K_H ratio, aquifer pore size multiplier, STOOIP, water relative permeability end point, residual oil saturation, critical water saturation, bottom of the perforated interval and applied drawdown. Examples of the extracted data are shown in the plots in Figure 9.

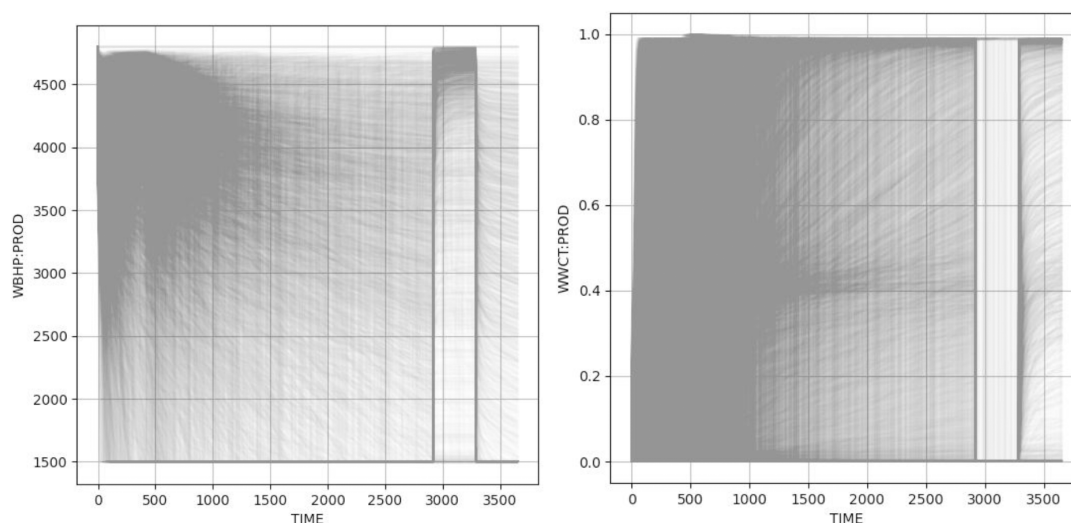


Figure 9—Sample of the extracted flowing bottomhole pressure (FBHP) and water cut for a number of simulation runs.

Variables normalization

To allow for a proper comparison of measured and calculated FBHP, it is necessary to normalize them. Several normalization methods were used, the most efficient resulted in the normalization used in pressure transient analysis; however, it opens the question of what single value of permeability to use to represent the heterogeneous cases. After considering several options, the one adopted was FBHP/P_i , P_i being the initial pressure of its drainage area. The derivative of the dimensionless pressure hints about the strength of the aquifer, which is the main energy source of all the cases considered in this paper.

In this study, both pressure and water-oil ratio (WOR) profiles from field data and conceptual numerical simulations were compared to identify characteristic reservoir behaviors through type curve matching. To ensure meaningful comparisons, it is necessary to normalize the data, as differences in absolute scale, due to reservoir size, pressure regime, or fluid properties, can mask the underlying dynamic patterns. Normalization aligns the datasets in terms of magnitude and time scale, allowing a shape-based comparison that is more robust to variations in initial conditions or measurement units.

An alternative approach explored in this work involved using the derivatives of pressure and WOR with respect to time, specifically on a logarithmic time scale. Derivative-based profiles emphasize the rate of change rather than absolute values, which can enhance sensitivity to flow regime transitions and displacement features. This approach inherently reduces the influence of differences in scale and baseline levels between datasets, allowing for more direct comparison of curve shapes across a range of reservoir conditions.

However, relying solely on derivatives does not fully eliminate the need for normalization. Derivatives are still sensitive to time, requiring interpolation or alignment to a common time basis, especially when comparing field data to synthetic curves generated on a different temporal space. Additionally, because numerical differentiation tends to amplify noise in field measurements, it was often found necessary to smooth out the signals before calculating the derivatives. To maintain consistency and ensure the comparability of all input features, normalization of both time and derivative values remains advisable, especially when using ML models that are sensitive to the scale of input data.

In summary, the use of derivatives enhances the ability to detect dynamic features and match the shape of pressure and WOR profiles, thereby improving the robustness of the type-curve matching process. Nevertheless, some degree of smoothing continues to be essential to facilitate automatic comparisons.

Features extraction

Following the simulation runs, a set of diagnostic features is computed to support the application of ML algorithms and aid in characterizing the underlying flow processes. The selected features are summarized in [Table 2](#).

Table 2—Features selected for the application of the machine learning (ML) algorithm

Water breakthrough	Time to WBT
	Np at WBT
Oil production rate	Peak oil production rate
Cumulative oil	at WBT time
	at 99% water cut or at the end of production
	at defined water cuts: 5% 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, and 95% water cut
Cumulative water	at the end of production
	at defined water cuts: 5% 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, and 95% water cut
Cumulative WOR	at defined water cuts: 5% 10%, 15%, 20%, 25%, and 30% water cut
Aquifer to STOOIP ratio	Initial water in place divided by the initial oil in place
Mobility ratio	at midpoint of normalized water saturation
Downhole pressure derivatives	after 1, 2 and 4 weeks of start up
	at the middle time between start-up and WBT
	before WBT
	at defined water cuts: 5% 10%, 15%, 20%, 25%, and 30% water cut
Water-Cut derivatives	at defined water cuts: 5% 10%, 15%, 20%, 25%, and 30% water cut

In addition to the previously described features, additional metrics were calculated to evaluate the well performance. These include the duration of each production stage, the estimated ultimate recovery (EUR), and the length of the production plateau. Notably, several of these metrics exhibit a measurable influence on EUR. For example, [Figure 10](#) illustrates the expected relationship between porosity and EUR: higher porosity leads to a greater volume of oil in place, which in turn results in higher ultimate recovery. While this outcome is expected, it serves as a useful validation point for the accuracy of the simulation results.

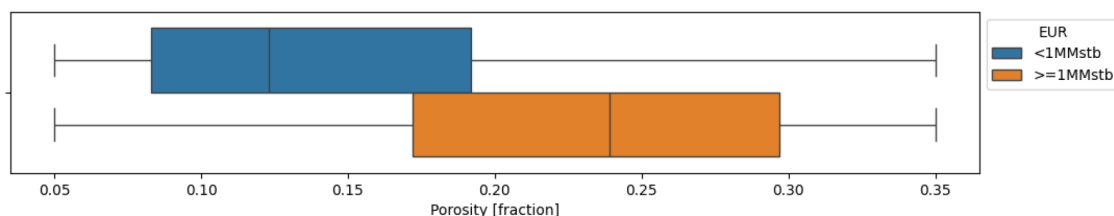


Figure 10—Relation between porosity and EUR. It indicates that the higher the porosity the larger the oil recovery

Other features reveal more insightful relationships. For instance, [Figure 11](#) shows a correlation between the length and position of the perforated interval and the duration of the production plateau. The results indicate that as the perforated interval approaches the original water-oil contact, the likelihood of bypassing the plateau phase increases. Conversely, a long production plateau is most likely when the perforation is in the upper half of the oil-bearing interval.

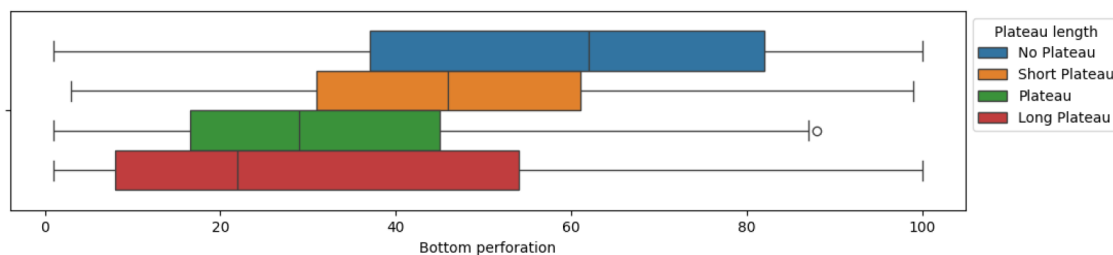


Figure 11—Relation between perforated length and length of the production plateau. Most favorable perforated intervals are within the upper half of the oil-bearing interval.

Features were calculated for both the pre-water breakthrough and post-breakthrough periods. The models were trained using the repeated cross-validation method⁷, stratified cross-validation in the case of categorical variables, considering 5 folds and 3 repeats. The performance of the models is evaluated using $f1_weighted$ ⁸ metric for the categorical variables and $r2_score$ ⁹ for the numerical variables.

Machine learning model training, testing and validation

For this study, training, testing, and validation are conducted separately depending on whether the well is producing dry oil or has already experienced water breakthrough. This means that the dataset is partitioned based on production stage, allowing the ML models to be trained and evaluated distinctly for pre-breakthrough and post-breakthrough conditions. This approach accounts for the following differences in flow behavior and feature reliability between these two production phases:

1. *before the water breakthrough*: during this production stage only pressure data is available and used for training. The same way, only pressure data will be used to predict.
2. *after the water breakthrough*: as soon as the water encroaches the well's completion interval the behavior of the FBHP can change, therefore, the pressure data is treated differently before and after the water breakthrough. Additionally, water production data is also available after the breakthrough.

The feature selection process and model algorithm selection are done using untuned models (hyperparameters set by default), allowing for faster experimentation, avoiding premature optimization, and giving a baseline performance suitable for comparison. Once the features and algorithm are chosen, the final model can be tuned to improve the performance of the predictions.

In this case, the selected algorithm is Random Forest, implemented in scikit-learn package for Python as *RandomForestRegressor*¹⁰ and *RandomForestClassifier*¹¹, and the tuned parameters are:

- the number of trees ($n_estimators$) increased to 500,
- the number of features to consider choosing the best split ($max_features$), set to a third of the total number of available features,
- Activating out-of-bag samples to estimate the generalization score, on the samples used to build the trees ($bootstrap=True$ and $oob_score=True$),

Tuning only these hyperparameters, we intend to improve the model performance without facing the risk of overfitting.

Feature analysis before water breakthrough

Before WBT, the reliability of the prediction of certain features is reduced, likely due to the relatively low cumulative production during this early phase. Nevertheless, key performance indicators such as estimated ultimate recovery (EUR) and production plateau duration can still be predicted with reasonable accuracy.

As shown in [Figure 12](#), these features achieve test scores of 0.8 or higher, demonstrating strong predictive performance despite the limited amount of production data available prior to WBT.

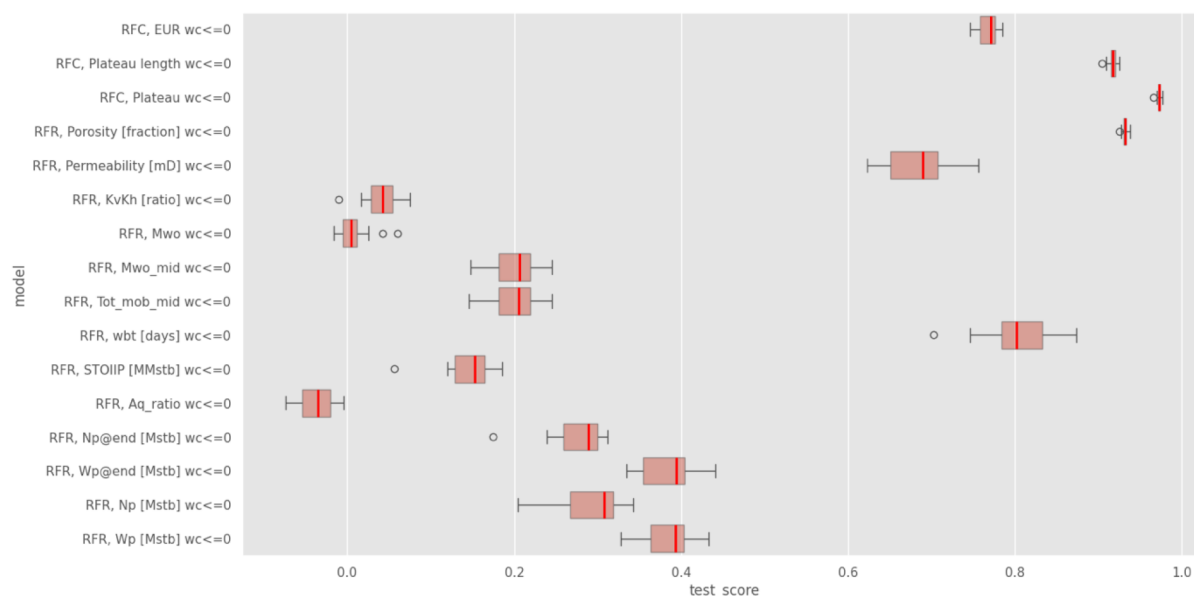


Figure 12—Model performance for the predicted features before water breakthrough (WBT)

Feature analysis after water breakthrough

Following WBT, the volume of available simulation data increases, enabling a better assessment of feature performance. As water cut progresses, the accumulation of production data leads to higher test scores, reflecting improved model reliability. This trend is clearly observed in the plots, where reservoir-related features shift to the right, indicating enhanced predictive strength. By the time the water cut reaches 30%, most features achieve reliability levels exceeding 75%, as illustrated in [Figure 13](#).

This marks one of the most critical outcomes of the methodology: at this stage, the model is capable of estimating reservoir characteristics with a high degree of confidence. Notably, the combined analysis of bottomhole pressure behavior and water cut evolution plays a central role in strengthening the reliability of these estimates.

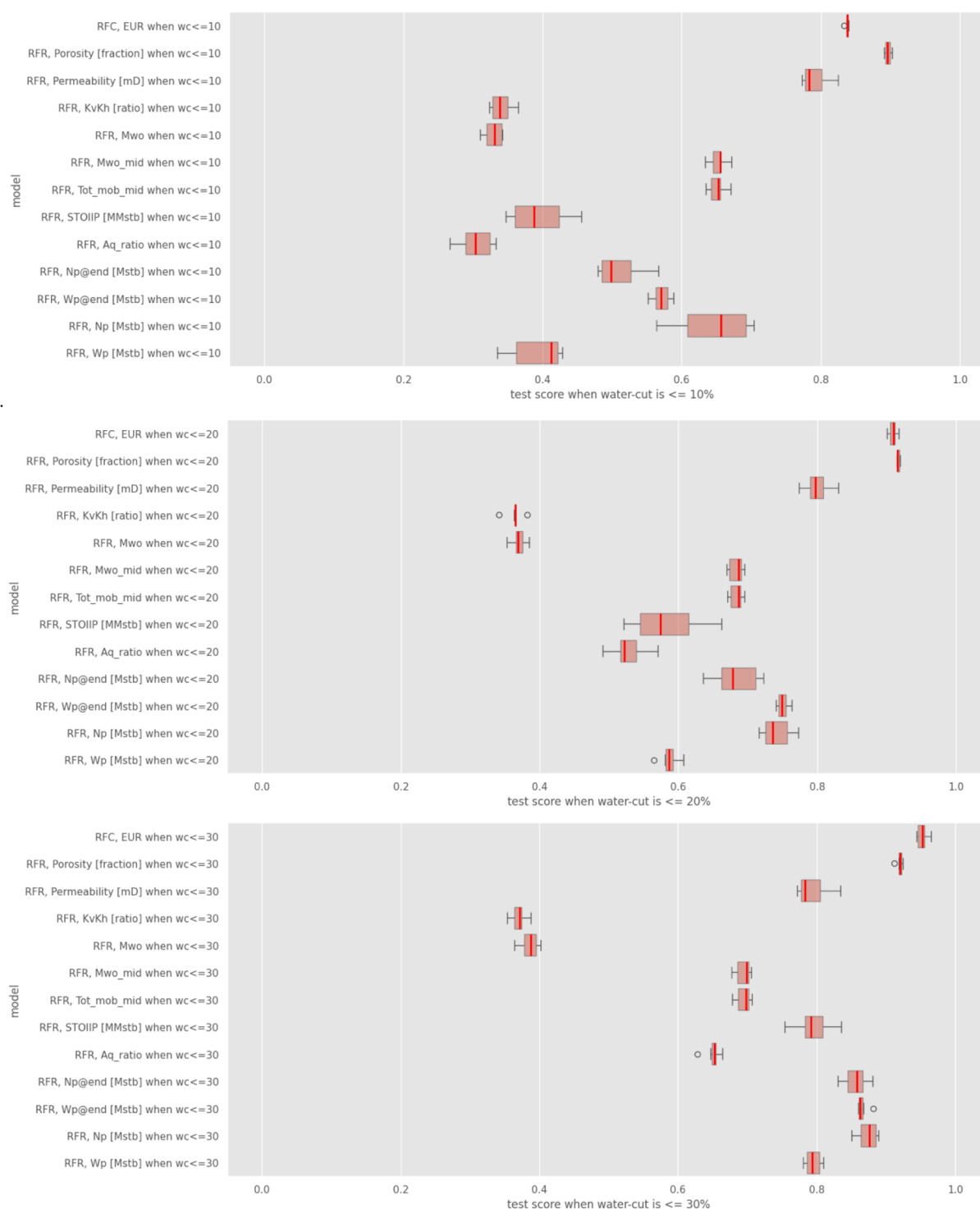


Figure 13—Model performance for the predicted features after water breakthrough (WBT)

Production forecasting using the trained models (before and after water breakthrough)

The trained models can be applied both to estimate key reservoir characteristics and to perform production forecasting. In the case study presented in this paper, the reservoir characteristics include: (1) rock properties such as porosity and permeability; (2) the contacted stock tank oil originally in place (STOOIP); (3) the size of the contacted aquifer relative to the contacted STOOIP; and (4) the water-oil mobility ratio.

Production forecasting can be performed using data collected either before or after WBT. Forecasts generated prior to WBT are inherently more uncertain due to the limited production data available at that stage. In contrast, forecasts made shortly after breakthrough, even when the water cut is still relatively low, tend to be more reliable and can provide valuable early insights for decision-making. To apply the machine learning model for production forecasting, the same set of input features used during training must be provided. These include the WBT time (if post-WBT data is used), as well as the time and cumulative oil derivatives of bottomhole pressure and water cut.

Finally, the corresponding model responds with the average estimated reservoir parameters and production profile, as can be seen in Figure 14. Taking advantage of the stochastic approach implemented by the random forest algorithm, the percentiles, like the most common P10, P50, and P90, for each one of the outputs can be also requested.

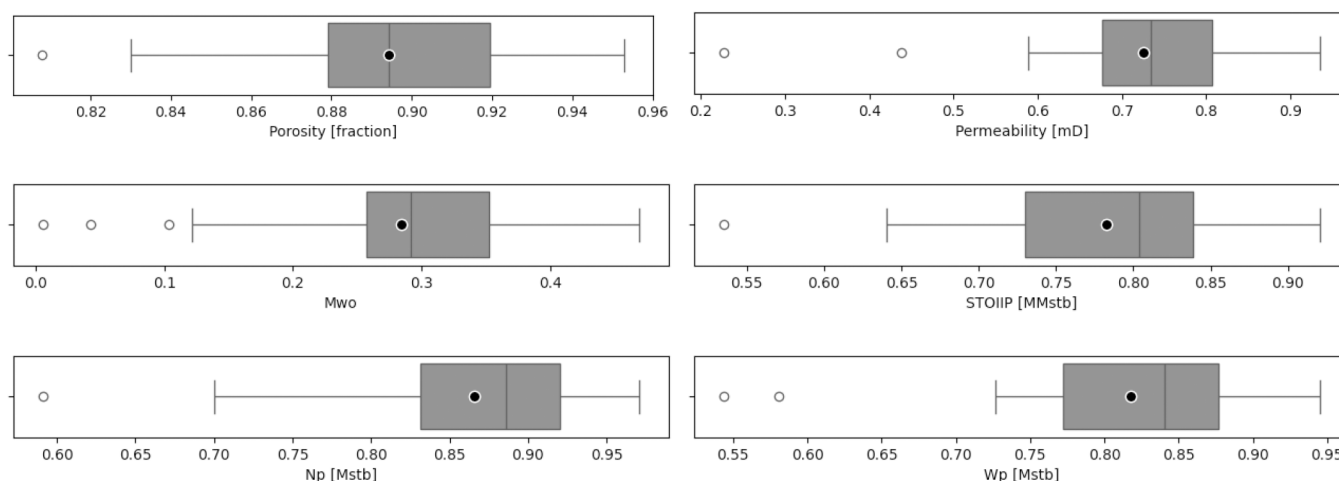


Figure 14—Tuned model performance for the predicted features after water breakthrough (WBT). The mean score is plotted as a black dot, on top of a standard boxplot.

Tuning Hyperparameters of Machine Learning Algorithm

As described before, only essential hyperparameters are tuned, with aim to maintain generalization and avoid overfitting, and after tuning the performance increases as expected without risking overfitting. These tuned models will be used to forecast the remaining production and to estimate the reservoir characteristics. The Figure 14 illustrates the r^2 score for the main output variables of the model; porosity, permeability, water-oil mobility ratio, contacted STOIP, oil cumulative and water cumulative; after tuning:

With the tuned model, the water-cut evolution vs. oil cumulative is plausibly forecasted, and a oil rate profile can be created to accompany the cumulative profile (Figure 15):

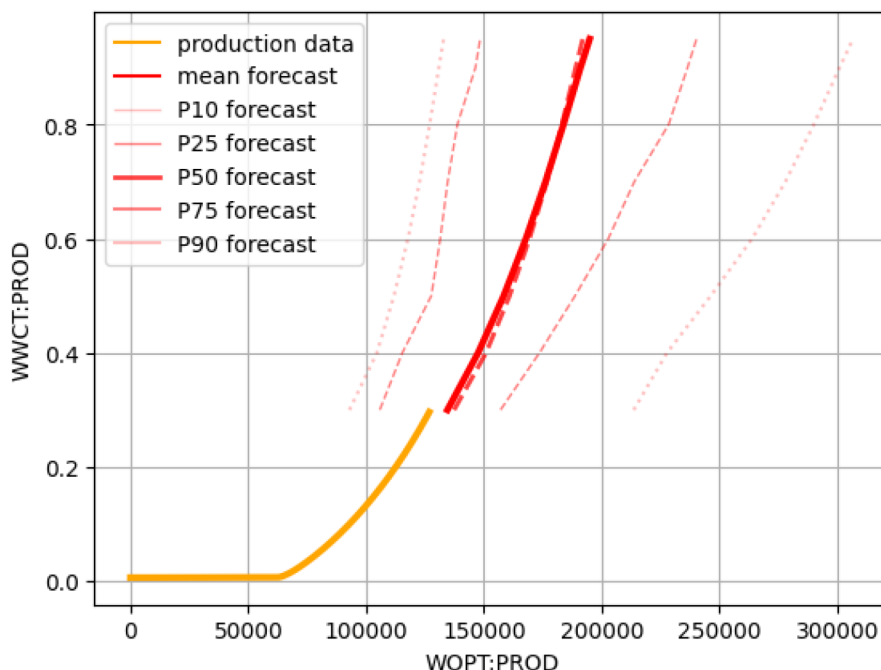


Figure 15—Model prediction of water-cut evolution. The chart shows the production data used for prediction in orange color, the average prediction profile as a solid red line, the P50 as dashed bold line, the P25 and P75 profiles as dashed thin lines, and the P10 and P90 as dotted thin lines.

In the same way, probabilistic estimations of porosity and permeability are also calculated by the trained model (Figure 16):

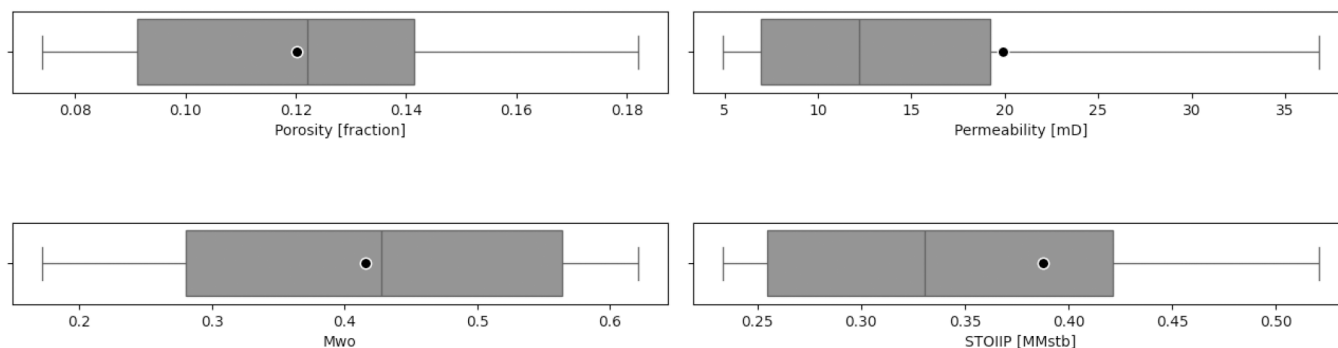


Figure 16—Model estimation for reservoir characteristics: porosity, permeability, water-oil mobility ratio, and STOIP.

The output of the model displays the expected value as a black point. Together with the mean value, a box plot shows the uncertainty range for each parameter, where the 50% of the possibilities lies within the gray rectangle (the box), the vertical line inside the box indicates the median (P50), and the whiskers limits the P10-P90 range.

The prediction from the trained model provides a valuable and swift outlook of the potential reserves from a well that has started to cut water. With that information, the reservoir engineers and management can promptly take decisions on the way to continue producing the well or remedial actions that could be implemented to reduce the water-cut.

The biggest advantage of this approach, compared to traditional reservoir simulation, relies on the easiness of the forecast setup and the speed of the simulations, allowing to convert production data into information and insights in a time effective manner.

Discussion

The application of WRM Process Modeling in this study demonstrates a robust, physics-informed approach to interpreting water production dynamics and bottomhole pressure behavior in heterogeneous reservoirs. By integrating synthetic simulations catalogs with ML techniques, the methodology reduces subjectivity in diagnosing aquifer connectivity, displacement patterns, and near-wellbore heterogeneity, key factors influencing water control strategies. The case study illustrates how this approach enables the identification of dominant geo-parameters affecting WOR and FBHP, thereby supporting data-driven decisions for remedial interventions.

A major strength of the WRM Process Modeling framework is its ability to bypass the computational burden and interpretive bias associated with traditional full-field simulation and history matching. Instead, it employs Monte Carlo sampling across a broad parameter space to generate a catalog of physically plausible conceptual models. This enables rapid matching between observed field data and synthetic signatures, allowing interpreters to identify the most probable reservoir configuration without relying on subjective judgment. The integration of random forest and neural network-based pattern recognition further enhances the objectivity and efficiency, particularly in capturing nonlinear interactions among parameters that are often missed by conventional regression or correlation analysis.

The normalization and derivative-based comparison of pressure and WOR profiles proved essential in aligning field and simulated data across different scales and temporal resolutions. While derivatives enhanced sensitivity to flow regime transitions and displacement features, they also introduced noise amplification, necessitating signal smoothing prior to matching. This underscores the importance of preprocessing steps in preserving the integrity of dynamic interpretations. Moreover, the normalization of FBHP using initial pressure (P_i) provided a dimensionless metric that facilitated comparisons across varying reservoir energy levels, offering insights into aquifer strength as a primary driver of water influx.

The implementation of vertical heterogeneity through rock type sequences derived from actual reservoir data significantly improved the realism of the conceptual models. The selection of favorable, average, and poor reservoir cases based on cumulative production metrics allowed for a stratified representation of geological variability. The use of heterogeneity indicators such as Shannon entropy, Markov transition entropy, and harmonic permeability enabled quantitative differentiation of rock type distributions across critical intervals: the original oil zone, the completion interval, and the lower hydrocarbon section. These indicators not only helped rank reservoir quality but also highlighted how vertical layering influences water coning and breakthrough timing.

An important implication of this work lies in its potential to inform targeted remedial actions. Once a conceptual model is matched to observed behavior, specific mitigation strategies, such as zonal isolation, conformance control, or inflow management, can be evaluated within the context of the identified reservoir setup. This avoids the inefficiencies of one-size-fits-all approaches and instead tailors interventions to the unique combination of geology, fluid mobility, and completion characteristics present in each well. Furthermore, the ensemble modeling approach provides uncertainty bounds around predictions, allowing operators to assess risk when selecting intervention options.

The integration of ML algorithms with physics-based simulations represents a paradigm shift in reservoir surveillance and water management. Unlike purely data-driven black-box models, which may lack physical consistency, or overly complex numerical simulators, which demand extensive calibration, the WRM Process Modeling approach offers a balanced middle ground. It combines the rigor of physics with the agility of ML-assisted diagnostics, resulting in a scalable and interpretable decision support tool. This is particularly valuable in environments where data scarcity or reservoir complexity limits the applicability of traditional analytical methods.

Finally, the environmental and operational relevance of this methodology cannot be overstated. As regulatory pressures mount on produced water volumes and disposal practices, efficient water control

becomes imperative for sustainable operations. By identifying the root causes of excessive water production, whether due to aquifer encroachment, fracture-driven channeling, or completion-induced inflow imbalance, the WRM Process Modeling workflow enables proactive, rather than reactive, management strategies. This contributes to both economic efficiency and environmental stewardship, aligning operational goals with broader sustainability objectives.

In conclusion, this study illustrates the effectiveness of combining domain expertise, synthetic simulation catalogs, and ML in unraveling complex reservoir behaviors. The WRM Process Modeling framework offers a structured, repeatable, and technically sound pathway for diagnosing water production issues and designing optimal mitigation strategies. Future enhancements could explore the integration of real-time production data and adaptive updating of the simulation catalog, further improving the responsiveness and predictive power of the approach.

Implementation of reservoir heterogeneity

The methodology has been demonstrated using a homogeneous reservoir case as an example. However, its greatest value lies in the incorporation of reservoir heterogeneity, which significantly enhances the realism and predictive capability of the model.

Heterogeneities can be effectively integrated into the WRM Process Modeling framework by replicating rock type distributions observed in actual reservoirs. Rock types are particularly valuable not only for their descriptive geological attributes but also because they incorporate fluid flow characteristics, enabling more accurate representation of reservoir dynamics.

To illustrate this approach, data from a specific reservoir was used to identify representative reservoir scenarios. Wells were ranked according to cumulative oil production and water production profiles over a defined period. For instance, wells in the top 5th percentile, characterized by high oil recovery and favorable water production behavior, were selected to define a "favorable" rock type sequence. Wells near the 50th percentile (P50) were used to represent average reservoir conditions, while those in the bottom 10th percentile were chosen to reflect lower-quality reservoir settings. The rock type patterns derived from these well groups provide a range of plausible heterogeneity scenarios that can be used in subsequent simulation and analysis.

For ML applications, it is more efficient to input quantified heterogeneity indicators rather than raw spatial property distributions. These indicators are used to characterize the variability of vertically stacked rock types in the near-wellbore region. Input data can be constructed using rock type sequences assigned to each well, typically under the assumption of a layer-cake geometry. This simplification focuses on vertical heterogeneity while neglecting lateral variations, which is often sufficient for many practical applications.

Several heterogeneity indicators have been evaluated, including:

- Conventional statistical parameters
- Shannon information entropy
- Markov transition entropy
- Rock type frequency change
- Harmonic permeability

Furthermore, these indicators are calculated across three distinct vertical intervals to improve resolution and relevance:

1. Initial oil-bearing interval: from the top of the reservoir to the depth of the original oil-water contact
2. Reservoir contact interval: corresponding to the perforated or open-hole section of the well

3. Lower hydrocarbon interval: from the base of the perforated or open-hole section down to the original oil-water contact

This multi-interval approach enables a more detailed characterization of vertical heterogeneity, which is crucial for assessing reservoir quality and predicting flow behavior. It also allows for the independent weighting of the influence that each interval may exert on water production dynamics, thereby improving the accuracy of diagnostic and predictive models.

Implications for Sustainable Operations

As regulatory oversight of produced water volumes and disposal practices continues to tighten, effective water control has become essential for maintaining both economic viability and environmental compliance. Excessive WOR not only increases operational costs but also poses significant environmental challenges, particularly in regions with stringent water management regulations.

WRM Process Modeling offers operators a technically robust and environmentally responsible approach to managing and reducing water production. By integrating physics-based reservoir simulation with ML techniques, the methodology enables data-driven decision-making that aligns operational strategies with sustainability objectives. It supports the rapid identification of root causes behind excessive water production and facilitates targeted remedial actions that enhance well performance while minimizing environmental impact.

Furthermore, WRM Process Modeling improves well intervention planning by providing a structured, repeatable workflow grounded in physical principles. This approach supports responsible field development under increasingly strict regulatory frameworks, ensuring that operational decisions are both economically sound and environmentally aligned.

Conclusions

- A new WRM Process Modeling approach enables fast, physics-based interpretation of WOR and FBHP data, providing actionable insights for reservoir management.
- The combination of conceptual numerical simulation with ML based parameter ranking significantly reduces interpretation subjectivity and enhances diagnostic accuracy.
- The methodology successfully identifies reservoir properties from FBHP and WOR performance. Prior to water breakthrough, when only pressure data is available, the reliability of the prediction of reservoir properties is noticeably less reliable. This limitation reflects the reduced information content in the pressure data alone.
- This workflow offers a practical solution for managing water production challenges in complex reservoirs without requiring full-field simulation or history matching.
- The integration of physics-based simulations with ML-driven pattern recognition improves both the technical rigor and operational efficiency of well intervention planning.
- WRM Process Modeling contributes to sustainable reservoir management by enabling informed, timely decisions that align with environmental and regulatory goals.

Nomenclature

BHP:	bottom hole pressure
EUR:	estimated ultimate oil recovery
FBHP:	flowing bottom hole pressure
N _p :	produced oil
ML:	machine learning
M _{ow} :	water-oil mobility ratio

P_i : initial reservoir pressure
SOWCR: critical water saturation (maximum S_w with $k_{rw}=0$)
STOOIP: stock tank oil originally in place
RFC: Random Forest Classifier
RFR: Random Forest Regressor
WBT: water breakthrough
WC: water-cut
WC&R: well completion and reservoir configurations
WOR: water-oil ratio
Wp: produced water
WRM: well and reservoir management